Subtracting Integers

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Integers

Integers are the whole numbers and their opposites.

$$\ldots, -2, -1, 0, 1, 2, \ldots$$

Subtraction

Subtraction can mean:

1. "taking away"
2. finding the **difference** between two numbers

Subtraction on a Number Line

On a **number line**:

Subtract a positive $\rightarrow$ move **left**
Subtract a negative $\rightarrow$ move **right**

Moving Left

Subtracting a positive integer moves us to the **left**.

$$5 - 3 = 2$$

Start at 5, move 3 units left, and land on 2.

Moving Right

Subtracting a negative integer moves us to the **right**.

$$5 - (-3) = 8$$

Start at 5, move 3 units right, and land on 8.

Opposites

The **opposite** (**additive inverse**) of a number has the same distance from zero but is on the other side.

opposite of 4 is -4

opposite of -7 is 7

The Rule

Subtracting an integer is the same as adding its **opposite**.

$$a - b = a + (-b)$$

Remember: Keep, Change, Change

Keep the first number.

Change subtraction to addition.

Change the second number to its opposite.

Keep, Change, Change

$$7 - 3$$

$$7 + (-3)$$

Keep 7, **change** $-$ to $+$, **change** 3 to -3.

Quick Guide

Subtraction	becomes	Addition
$8-3$	$\rightarrow$	$8+(-3)$
$8-(-3)$	$\rightarrow$	$8+3$

Two Negatives Make a Positive

Subtracting a negative is the same as adding a positive.

$$a - (-b) = a + b$$

$$4 - (-6) = 4 + 6 = 10$$

Examples

Example 1

Find $7 - 3$.

$$7 - 3 = 7 + (-3) = 4$$

Example 2

Find $1 - 5$.

$$1 - 5 = 1 + (-5) = -4$$

Example 3

Find $5 - (-2)$.

$$5 - (-2) = 5 + 2 = 7$$

Example 4

Find $-3-(-8)$.

$$-3-(-8)=-3+8=5$$

Example 5

Find $-6 - 4$.

$$-6 - 4 = -6 + (-4) = -10$$

Practice Exercises

Practice 1

Solve.

$$6 - 2$$

Practice 2

Solve.

$$3 - 7$$

Practice 3

Solve.

$$-5 - (-2)$$

Practice 4

Solve.

$$-3 - 4$$

Practice 5

Solve.

$$2 - 6$$

Practice 6

Solve.

$$-3-(-5)$$

Practice 7

Solve.

$$4 - (-3)$$

Practice 8

Solve.

$$-12 - 8$$

Practice 9

Solve.

$$-15 - (-9)$$

Practice 10

Solve.

$$3 - (-5)$$

Activities

Activity 1

Solve.

$$9 - 5$$

Activity 2

Solve.

$$4 - 8$$

Activity 3

Solve.

$$-6 - (-3)$$

Activity 4

Solve.

$$-2 - 5$$

Activity 5

Solve.

$$3 - 8$$

Activity 6

Solve.

$$-2 - (-6)$$

Activity 7

Solve.

$$5 - (-4)$$

Activity 8

Solve.

$$-14 - 9$$

Activity 9

Solve.

$$-18 - (-11)$$

Activity 10

Solve.

$$2 - (-7)$$

Thank you!